

Specification Packaging Material

E-63140-00, LEAFLET GRANODEX INJECTION MALAYSIA

Author(s):

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Reviewed by:

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Approved by:

12.11.2021, RA Officer 02, Regulatory Affairs Officer,

180 mm

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Granodex 1 mg/ml Solution for Injection or Infusion



Composition:

Each ml contains
Granisetron HCl equivalent to Granisetron 1 mg

Excipients in Granodex 1 mg/ml solution for injection or infusion:
Sodium chloride
Citric acid monohydrate
Sodium hydroxide
Water for injection

Description:

Clear, colorless solution.
Clear and colorless solution after dilution.

Pharmacodynamics:

Serotonin receptors of the 5-HT₃ type are located peripherally in vagal nerve terminals and centrally in the chemoreceptor trigger zone of the area postrema. During chemotherapy-induced vomiting, mucosal enterochromaffin cells release serotonin, which stimulates 5-HT₃ receptors. This invokes vagal afferent discharge, inducing vomiting. Granisetron is a potent antiemetic and highly selective antagonist of 5-hydroxytryptamine (5-HT₃) receptors. Granisetron has negligible affinity for other receptor types including 5-HT and dopamine D₂ binding site.

Pharmacokinetics:

Absorption

Absorption of granisetron is rapid and complete, though oral bioavailability is reduced to about 60% as a result of first pass metabolism.

Distribution

Granisetron is extensively distributed, with a mean volume of distribution of approximately 3l/kg. Plasma protein binding is approximately 65%.

Metabolism

Biotransformation pathways involve N-demethylation and aromatic ring oxidation followed by conjugation. Granisetron's major route of metabolism is inhibited by ketoconazole, suggestive of metabolism mediated by the cytochrome P450 3A subfamily.

Elimination

Clearance is predominantly by hepatic metabolism. Urinary excretion of unchanged granisetron averages 12% of dose while that of metabolites amounts to about 47% of dose. The remainder is excreted in feces as metabolites. Mean plasma half-life in patients by the oral and intravenous route is approximately 9 hours.

Special Populations:

Renal failure

In patients with severe renal failure, data indicate that pharmacokinetic parameters after a single intravenous dose are generally similar to those in normal subjects.

Hepatic impairment

In patients with hepatic impairment due to neoplastic liver involvement, total plasma clearance of an intravenous dose was approximately halved compared to patients without hepatic involvement. Despite these changes, no dosage adjustment is necessary.

Elderly

In elderly subjects after single intravenous doses, pharmacokinetics parameters were within the range found for non-elderly subjects.

Paediatrics

In children, after single intravenous doses, pharmacokinetics are similar to those in adults when appropriate parameters (volume of distribution, total plasma clearance) are normalized for body weight.

Indications:

Granisetron is indicated for the prevention and treatment (control) of:
- acute and delayed nausea and vomiting associated with chemotherapy and radiotherapy
- postoperative nausea and vomiting

Dosage and directions for use:

Chemotherapy induced nausea and vomiting (CINV)

Adults

Prevention:

A dose of 1-3 mg (10-40 mcg/kg) of granisetron should be administered either as a slow intravenous injection (over 30 seconds) or as an intravenous infusion diluted in 20 to 50 ml infusion fluid and administered over 5 minutes, prior to the start of chemotherapy.

Treatment:

A dose of 1-3 mg (10-40 mcg/kg) granisetron should be administered either as a slow intravenous injection (over 30 seconds) or as an

intravenous infusion diluted in 20 to 50 ml infusion fluid and administered over 5 minutes. Further treatment doses of granisetron may be administered, if required, at least 10 minutes apart. The maximum dose of granisetron to be administered over 24 hours should not exceed 9 mg.

Paediatrics

Prevention and treatment:

A dose of 10-40 mcg/kg body weight (up to 3 mg) should be administered as an intravenous infusion, diluted in 10 to 30 ml infusion fluid and administered over 5 minutes prior to the start of chemotherapy. One additional dose may be administered within a 24 hours period if required. This additional dose should not be administered until at least 10 minutes after the initial infusion.

Radiotherapy induced nausea and vomiting (RINV)

Adults

Prevention:

A dose of 1-3 mg (10-40 mcg/kg) of granisetron should be administered either as a slow intravenous injection (over 30 seconds) or as an intravenous infusion diluted in 20 to 50 ml infusion fluid and administered over 5 minutes, prior to the start of radiotherapy.

Treatment:

There is insufficient information to recommend the intravenous administration of granisetron in the treatment of RINV in adult patients.

Paediatrics

There is insufficient information to recommend intravenous administration of granisetron in the prevention and treatment of RINV in children.

Postoperative nausea and vomiting (PONV)

Adults

Prevention:

A dose of 1 mg (10 mcg/kg) of granisetron should be administered as a slow intravenous injection (over 30 seconds) prior to induction of anesthesia.

Treatment:

A dose of 1 mg (10 mcg/kg) of granisetron should be administered by slow intravenous injection (over 30 seconds). The maximum dose for patients undergoing anesthesia for surgery is a total dose of 3 mg of granisetron intravenous in one day.

Paediatrics

There is insufficient information to recommend intravenous administration of granisetron in the prevention and treatment of postoperative nausea and vomiting in paediatric patients.

Special dosage instruction:

Elderly

: no dosage adjustment required.

Renal impairment

: no dosage adjustment required.

Hepatic impairment

: no dosage adjustment required.

Instructions for use, handling and disposal

Preparation of infusion

For adults:

The appropriate dose is diluted with a compatible infusion fluid to a total volume of 20 to 50 ml, in any of the following solutions: 0.9% sodium chloride, Ringer's lactate, 5% dextrose and 20% mannitol solution. The mixtures solutions are stable within 24 hours at temperatures below 30°C.

For children:

The appropriate dose is diluted with a compatible infusion fluid (as for adults) to a total volume of 10 to 30 ml, in any of the following solutions: 0.9% sodium chloride, Ringer's lactate, 5% dextrose and 20% mannitol solution. The mixtures solutions are stable within 24 hours at temperatures below 30°C.

From a microbiological point of view the diluted product should be used immediately. If not used immediately, in use storage times and conditions prior to use are the responsibility of the user and the mixtures solutions are stable within 24 hours at temperatures below 30°C.

Incompatibilities:

Admixtures of granisetron hydrochloride and dexamethasone sodium phosphate are compatible at concentrations of 10 to 60 mcg/ml granisetron and 80 to 480 mcg/ml dexamethasone phosphate in either 0.9% sodium chloride or 5% glucose intravenous infusion fluids.

Route of Administration:

Intravenous injection or infusion.

Contraindications:

Granisetron is contraindicated in patients with known hypersensitivity to granisetron or to any of its excipients.

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Last revised: dd/mm/yyyy

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Fiber direction

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Page 1

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Warnings and Precautions:

- As granisetron may reduce lower bowel motility, patients with signs of subacute intestinal obstruction should be monitored closely following administration of granisetron.
- As with other 5-HT₃ antagonists, cases of ECG modifications including QT prolongation have been reported with granisetron. These ECG changes with granisetron were minor and generally not of clinical significance, specifically with no evidence of proarrhythmia. However, in patients with pre-existing arrhythmias or cardiac conduction disorders, this might lead to clinical consequences. Therefore, caution should be exercised in patients with cardiac co-morbidities, on cardio-toxic chemotherapy and/or with concomitant electrolyte abnormalities. Cross-sensitivity between 5-HT₃ antagonists has been reported.
- As with other 5-HT₃ antagonists, cases of serotonin syndrome (including altered mental status, autonomic dysfunction and neuromuscular abnormalities) have been reported following the concomitant use of granisetron and other serotonergic drugs. If concomitant treatment with granisetron and other serotonergic drugs is clinically warranted, appropriate observation of this patient is advised.

Effects on ability to drive and use machines

Granisetron has no or negligible influence on the ability to drive or to use machines.

Interactions with Other Medicaments:

- Hepatic enzyme induction with phenobarbital resulted in an increase in total plasma clearance of intravenous granisetron of approximately one-quarter.
- Ketoconazole inhibited ring oxidation of granisetron. However, given the absence of pK/pD relationship with granisetron, these changes are believed to have no clinical consequences.
- Granisetron has been safely administered in humans with benzodiazepines, neuroleptics and antiulcer medications, commonly prescribed with antiemetic treatments. Additionally, granisetron has shown no apparent drug interaction with emetogenic cancer chemotherapies.
- No specific interaction studies have been conducted in anesthetized patients, but granisetron has been safely administered with commonly used anesthetic and analgesic agents. In addition, the activity of the cytochrome P450 subfamily 3A4 (involved in the metabolism of some of the main narcotic analgesic agents) is not modified by granisetron.
- As for other 5-HT₃ antagonists, cases of ECG modifications including QT prolongation have been reported with granisetron. These ECG changes with granisetron were minor and generally not of clinical significance, specifically with no evidence of proarrhythmia. However, in patients concurrently treated with drugs known to prolong QT interval and/or are arrhythmogenic, this may lead to clinical consequences.
- As with other 5-HT₃ antagonists, cases of serotonin syndrome have been reported following the concomitant use of granisetron and other serotonergic drugs. If concomitant treatment with granisetron and other serotonergic drugs is clinically warranted, appropriate observation of this patient is advised.

Pregnancy and Lactation:

There are no studies in pregnant women and it is not known whether granisetron is excreted in human milk. Use of granisetron during pregnancy or lactation should be limited to situations where the potential benefit to the mother justifies the potential risk to the fetus or nursing infant.

Adverse Effects:

The most frequently reported adverse reactions for granisetron are headache and constipation which may be transient. ECG changes including QT prolongation have been reported with granisetron. The following table of listed adverse reactions data associated with granisetron.

<i>Immune system disorders</i>	
Uncommon	Hypersensitivity reactions e.g. anaphylaxis, urticaria
<i>Nervous system disorders</i>	
Very common	Headache
Uncommon	Serotonin Syndrome
<i>Cardiac disorders</i>	
Uncommon	QT prolongation
<i>Gastrointestinal disorders</i>	
Very common	Constipation
<i>Hepatobiliary disorders</i>	
Common	Elevated hepatic transaminases
<i>Skin and subcutaneous tissue disorders</i>	
Uncommon	Rash

Overdose and Treatment:

There is no specific antidote for granisetron. In the case of overdosage with granisetron, symptomatic treatment should be given. Overdosage of up to 38.5 mg of granisetron hydrochloride as a single injection has been reported without symptoms or only the occurrence of a slight headache.

Shelf-life:

Granisetron 1 mg/ml solution for injection or infusion can be used within 24 months from the date of manufacturer if kept as recommended (see **Storage Conditions**).

Presentations and Registration Numbers:

GRANODEX 1 mg/ml : Box, 5 ampoules x 1 ml; MALXXXXXXXXXAZ
GRANODEX 3 mg/3 ml: Box, 5 ampoules x 3 ml; MALXXXXXXXXXAZ

ON MEDICAL PRESCRIPTION ONLY.**Storage Conditions:****Unopened ampoules**

STORE AT TEMPERATURES BELOW 30°C.

KEEP THE AMPOULES IN THE OUTER CARTON. PROTECT FROM LIGHT.

After dilution

Granodex 1 mg/ml solution for injection or infusion is demonstrated to be physically and chemically stable for 24 hours when stored at temperatures below 30°C when diluted with 0.9% NaCl, Ringer's lactate, 5% Dextrose, and 20% Mannitol.

From a microbiological point of view the diluted product should be used immediately. If not used immediately, in use storage times and conditions prior to use are the responsibility of the user and the mixtures solutions are stable within 24 hours at temperatures below 30°C.

Manufactured by**PT Ferron Par Pharmaceuticals**

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 Jl. Jababeka VI Blok J3, Cikarang
 Bekasi-Indonesia

For**PT Dexa Medica**

Jl. Jend. Bambang Utoyo No. 138
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Product Registration Holder**Averroes Pharmaceuticals Sdn. Bhd**

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Date of revision of package insert: 08 September 2021

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Fiber direction

No.	Parameters	Specification
1.	Material	HVS (Lihat CoA / See CoA)
2.	Gramatur / Gramature	54 – 66 gsm
3.	Dimensi panjang / Length dimension	239 – 241 mm
4.	Dimensi lebar / Width dimension	179 – 181 mm
5.	Arah serat / Fiber direction	Vertikal / Vertical
6.	Warna / Colour	Lihat proof print / See proof print
7.	Redaksional / Redactional	Lihat proof print / See proof print

HISTORY

Revision No.	Revision Date	Details	Reason
00	-	Dokumen baru menggantikan Specpacgen006668/1, dengan perubahan : - Perubahan bentuk sediaan menjadi "Solution for injection or infusion" - Perubahan redaksional sesuai PI terbaru	CMCOthRec075034/1
01	10 Sep 2021	Penghilangan redaksional mengikuti PI terakhir	- CMCOthRec075035/2 - CMCOthRec077297/1
02	11 Nov 2021	- Perubahan ukuran font redaksional menjadi 7pt - Penambahan kotak keterangan ukuran dan jenis font yang digunakan pada redaksional	CMCOthRec079634/1